

International Journal of Developmental Disabilities

ISSN: 2047-3869 (Print) 2047-3877 (Online) Journal homepage: https://www.tandfonline.com/loi/yjdd20

The effects of the class-wide function-related intervention teams (CW-FIT) on behaviors of an elementary student with autism spectrum disorder (ASD) in an inclusive classroom in Taiwan

Yi-Chen Wu, Pei-Yu Chen, Shin-Ping Tsai, Shu-Fei Tsai, Yu-Chi Chou & Chun-Yu Chiu

To cite this article: Yi-Chen Wu, Pei-Yu Chen, Shin-Ping Tsai, Shu-Fei Tsai, Yu-Chi Chou & Chun-Yu Chiu (2019): The effects of the class-wide function-related intervention teams (CW-FIT) on behaviors of an elementary student with autism spectrum disorder (ASD) in an inclusive classroom in Taiwan, International Journal of Developmental Disabilities, DOI: 10.1080/20473869.2019.1647031

To link to this article: https://doi.org/10.1080/20473869.2019.1647031

Published online: 06 Aug 2019.

Submit your article to this journal

Article views: 16

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=yjdd20

Check for updates

The effects of the class-wide function-related intervention teams (CW-FIT) on behaviors of an elementary student with autism spectrum disorder (ASD) in an inclusive classroom in Taiwan

Yi-Chen Wu[1], Pei-Yu Chen[2], Shin-Ping Tsai[2], Shu-Fei Tsai[3], Yu-Chi Chou[4] and Chun-Yu Chiu[5]

[1]Wenhua Elementary School, Taipei, Taiwan; [2]National Taipei University of Education, Taipei, Taiwan; [3]National University of Tainan, Tainan, Taiwan; [4]Chung Yuan Christian University, Taoyuan City, Taiwan; [5]National Taiwan Normal University, Taipei, Taiwan

In Taiwan, most students with disabilities receive education in an inclusive setting. Literature has documented the effects of interventions in increasing students' positive behaviors in inclusive settings, including students with disabilities in Western countries; however, effectiveness of such interventions in an Asian context remains unclear. The Class-Wide Function-Related Intervention Teams (CW-FIT) is one of the interventions that applies reinforcement-based strategies and provides multi-tiered supports to students with various severity of challenging behaviors. This study investigates the effects of CW-FIT Tier I (i.e. class-wide intervention) and Tier II (i.e. self-management) on the on-task and disruptive behaviors of a student with Autism Spectrum Disorder in an inclusive elementary classroom in Taiwan. Across nine weeks of intervention, the researchers used a reversal single-case design A-B-C-B-C to demonstrate experimental control over five phases. In addition, the researchers administered interviews and questionnaires to collect social validity data from the teacher and peers' perceptions toward the intervention. Findings from this study support that the CW-FIT is an effective intervention in increasing a student's on-task behaviors and decreasing disruptive behaviors in an inclusive classroom in an Asian context. The effect of implementing multiple tiers of CW-FIT was much more effective than implementing solely Tier I (class-wide intervention). The intervention was also well-received by the general education teacher and students.

Keywords: behavioral intervention, positive behavior support, inclusive education, autism

Inclusive education began in western society as an initiative to ensure access to general education for students with disabilities, and it has become a global trend due to its documented benefits for all stakeholders (Ferguson 2008). In the Republic of China (Taiwan), the majority of the primary and secondary students with disabilities receive education in education settings that are inclusive ($n = 87{,}183$, 82.5%; Ministry of Education 2017). The statistics seemed to validate that Taiwan has followed the movement to provide inclusive education to students with and without disabilities. However, inclusive education is more than an educational placement and should include providing environmental and pedagogical modification for all students to overcome barriers to learn (UN Committee on the Rights of Persons with Disabilities 2016). To achieve this goal, it is pertinent to prepare educators to implement inclusion in their classrooms (Forlin and Chambers 2011).

Unfortunately, past studies in Taiwan resonated with western literature, showing that general education teachers sometimes express refusal toward inclusion as a result from insufficient supports (Chiu 2001, Pivik *et al.* 2002, De Boer *et al.* 2011, Chen and Chiu 2015). Most general education teachers still reported concerns about providing appropriate services to students with disabilities due to their limited training (Lin 2011). Among all students with disabilities, general education teachers report students with autism spectrum disorder

Correspondence to: Chun-Yu Chiu, National Taiwan Normal University, Taipei, Taiwan. Email: chiuc@ntnu.edu.tw

© The British Society of Developmental Disabilities 2019
DOI 10.1080/20473869.2019.1647031

(ASD) with more challenges for classroom management as compared to their peers (Huang and Lan 2012). In addition to common classroom management strategies, many general education teachers in Taiwan provide accommodations for students with disabilities, including ASD. Such accommodations include assigning appropriate seats, using peers or visual cues to prompt positive behaviors, applying a point system, providing priming, and allowing more time to complete assignments and tests (Chang and Lin 2011, Sun and Huang 2018). Many of the strategies, however, may not meet the specific needs of the students with disabilities in their class (Sun and Huang 2018). General education teachers find it difficult to accommodate each student and to address behavioral issues with a disproportionate student–teacher ratio, especially because teachers are obligated to keep their teaching with an examination-driven curriculum on schedule (Hsu and Chan 2008). In addition to inadequate preparation, the lack of resources (including time to collaborate and consult special education teachers, teaching materials, support from administrators) poses threats on high quality teaching, which may lead to undesirable classroom atmosphere, student behavioral and academic outcomes (Chiang 2011).

Researchers have developed and validated interventions in inclusive classrooms, aiming to increase students' on-task behaviors as well as decrease their disruptive behaviors. Class-wide Function-related Intervention Teams (CW-FIT; Wills *et al.* 2010) is one of the interventions grounded in positive behavior support principles which emphasizes on reinforcement-based strategies. The principles of positive behavior support align well with education policy in Taiwan, which emphasizes on fostering positive atmosphere in school settings (Educational Fundamental Act 2013, Enforcement Rules of the Special Education Act 2013).

Class-wide function-related intervention teams (CW-FIT)

General education teachers need strategies to engage all students in their classes. CW-FIT provides behavior intervention strategies to promote effective classroom management, in which it employs applied behavior analysis strategies and embeds game into teachers' instructional activities (Kamps *et al.* 2011). CW-FIT is a multi-tiered intervention that allows teachers to provide a continuum of supports contingent upon severity of students' challenging behaviors.

CW-FIT is designed to be implemented by general education teachers, who teach and manage behavior problems simultaneously. Wills *et al.* (2010) recommended teachers to adopt class-wide intervention when more than 20% of students become off-task, and to implement individualized interventions if only a few students are off-task. CW-FIT Tier I intervention (hereafter referred to as Tier I) contains direct instructions on appropriate classroom behaviors with group contingency to reinforce appropriate behavior while decreasing inappropriate behaviors. Research confirms that when teachers establish and implement classroom rules consistently, students' disruptive behaviors decrease significantly (Marzano 2013). By design, the teacher instructs three behavior expectations to students by defining, modeling and role-playing in Tier I. These three behavior expectations include (a) getting teacher's attention properly, (b) following directions immediately, and (c) ignoring peers' inappropriate behaviors (CW-FIT n.d.). Each skill includes three to four steps. For example, the steps for getting teacher's attention properly are looking at teacher, raising your hand, waiting for teacher to call on you, and asking a question/giving your answer (CW-FIT n.d.). Then the teacher divides students into teams and reinforces the three behavior expectations using a group contingency strategy. The teacher, using a timer to prompt him/herself to give feedback every 2–5 min, awards each team with points and positive feedback. Upon reaching the pre-set goal, the team can exchange points earned for a reward (Wills *et al.* 2010).

When there are students who fail to respond to Tier I, the teacher would introduce CW-FIT Tier II intervention (hereafter referred to as Tier II) to this group of students. Specifically, the teacher teaches these target students self-management strategies by using a self-management chart (i.e. visual prompt for the three behavior expectations) and auditory cues. When the set interval of time is up, the teacher provides feedback to each team and prompts the target students to give themselves a point if they meet the three behavior expectations (Wills *et al.* 2010).

Since 2010, an increasing number of literature has documented the evidence supporting the effectiveness and feasibility of CW-FIT in inclusive classrooms in the United States. It is evident that CW-FIT can improve on-task behaviors across grade levels from preschool (Jolstead *et al.* 2017), elementary school (Wills *et al.* 2010, Kamps *et al.* 2011, Caldarella *et al.* 2015) to junior high school (Conklin *et al.* 2017). Moreover, changes of on-task behaviors for whole class and individual students are evident in two randomized control trials (Wills *et al.* 2016, 2018). In addition, Conklin *et al.* (2017) measured observable behaviors directly related to the three behavior expectations (i.e. talking out, compliance, and hand-raising) and found that CW-FIT had a desirable effect on these related behaviors.

Findings from past studies have also shown that CW-FIT is effective in increasing students on-task behaviors and decreasing disruptive behaviors for individual students with diverse needs: at-risk students or students with serious behavior disorders (Wills *et al.* 2010, Kamps *et al.* 2011, Caldarella *et al.* 2015), students with learning disability and hearing impairment

(Wills *et al.* 2014), and English language learning students with behavior problems (Naylor *et al.* 2018). While literature confirmed the effects of CW-FIT on students with a range of disabilities, limited data is available on students with ASD and thus require further investigation.

Although past studies have validated effects of CW-FIT in the United States, there remained a lack of research on effectiveness of CW-FIT in inclusive classrooms in an Asian context, despite the known fact that the classroom experience can be very different from a Western country. The differences, ranging from class size, student interaction, teaching approach, class activity/student involvement and teacher authority, bring out questions of effectiveness of interventions in a non-Western context (Aldridge *et al.* 1999, Lai *et al.* 2015). Furthermore, most of the CW-FIT studies were on at-risk students or students with serious behavior disorder, it is unclear how well CW-FIT would impact students with ASD in Taiwan. Therefore, the purpose of this study is to examine the effects and social validity of CW-FIT on the behaviors of an elementary student with ASD. Specifically, the research questions are as follows:

a. Does the implementation of CW-FIT result in increased on-task behavior for a student with ASD in an inclusive classroom?
b. Does the implementation of CW-FIT result in decreased disruptive behaviors for a student with ASD in an inclusive classroom?
c. Do the general education teacher and other students perceive CW-FIT as socially valid?

Method

Participants and setting

The research team recruited students and teachers from a fifth-grade general education classroom in a suburb area in Northern Taiwan. All 23 students in the classroom received Tier I intervention (class-wide intervention) and three target students (including a student with ASD) received Tier II intervention (self-management) from the general education teacher who was the homeroom teacher of the selected class. The general education teacher had more than 10 years of teaching experience and has taught students with disabilities before. The first author was the resource room teacher at school with 5 years of teaching experiences. She received CW-FIT training from the second author, provided support to the general education teacher, and collected data from the site.

The general education teacher referred a student, Ted, to the first author to participate in this study because of his low engagement during Chinese literacy classes. The target student, Ted, was a 10-year-old boy with an ASD diagnosis. Ted spent most of his school day in the general education class. In addition to the literacy and math classes taught by the general education teacher, Ted received pull-out service in resource classroom four hours per week, usually early in the morning or during lunch recess, for additional Chinese literacy and math instructions. Ted's IQ, assessed by Wechsler Preschool and Primary Scale of Intelligence-Third Edition, was 78 (verbal IQ = 84; performance IQ = 71). He used short phrases to express needs and to answer questions but he could neither express his feelings nor explain why he was upset. The general education teacher sought assistance from the first author because Ted would play with objects (e.g. pencils, water bottle) and seldom follow directions or complete assignments during class period, even with prompts and physical assistance. Ted had difficulties staying on-task in the general education class, despite of the fact that he had no problems following directions in the resource classroom.

Research design

The researchers used a reversal single-case design A-B-C-B-C to demonstrate experimental control in the intervention (Kennedy 2005), including phases as follows: baseline (Phase A), Tier I (Phase B_1 and B_2), and Tier I + Tier II (Phase C_1 and C_2). The first author conducted the experiment during Chinese literacy class every Tuesday and Thursday for nine weeks and collected 29 data points. We explain each phase below.

Baseline (A): Baseline phase was "business as usual" in Chinese literacy class; including a 2-min meditation (i.e. students sitting quietly on their seats with eyes closed), reading aloud, vocabulary instruction, team discussion, and assignments. Students were assigned to six teams with three to four students in each team. A reinforcement system was in place during baseline. When the teacher saw a student's good behavior (e.g. answer questions or on-task), she would award a token for the student's team. The team with most tokens at the end of the week could earn a tea party with the teacher on Friday as a reward.

CW-FIT Tier I (B_1 and B_2): The general education teacher implemented Tier I to all students during Chinese Literacy class every Tuesday and Thursday. At the beginning of each class, the general education teacher reviewed three behavior expectations with the students and set goal of points for the whole class. During the first 2 weeks of intervention, the timer was set for a 3-min interval and then increased to 5-min to remind the general education teacher to give team behavioral feedback. When the time was up, the general education teacher gave a point to each team meeting the behavior expectations along with specific verbal feedback. When a team member failed to show appropriate behaviors during the time period, the teacher reminded and encouraged the team verbally to earn a point at the end of the next time-period.

At the end of each class session, the teams meeting the set goal earned a token and could exchange tokens for the reward on the reinforcer menu. When Ted's behavior reached stability or the trend showed a decreasing trend for on-task behavior for more than three consecutive data points, we moved on to implementing the next phase.

CW-FIT Tier I + Tier II (C_1 and C_2): During these phases, at the beginning of the class, the general education teacher reviewed the rules of Tier I and introduced Tier II to the whole class. Next, the general education teacher selected Ted and two typical-developing peers from other teams who are considered as good role models of the class as target students for Tier II. The general education teacher then implemented Tier I intervention to all students, with the three target students (i.e. Ted and two typical-developing peers) conducting additional self-management (Tier II). For the whole class, the team used the same reward system as phase B. The three target students each got an individual point chart to put on their desk. At the end of each 5-min interval, the general education teacher gave each team behavioral feedback; at the same time she praised the three target students for their positive behaviors and asked each of the students to give themselves a point on their chart upon meeting the three behavior expectations.

When Ted's on-task behavior reached stability for three data points during phase C_1, the researchers stopped Tier II and return to Tier I only (i.e. phase B_2). The researchers stopped the experiment session once the record showed stable on-task behavior for at least three data points during phase C_2.

Research procedures

Teacher training and student instruction: Tier I: Prior to baseline, the first author used materials from the CW-FIT website to train the general education teacher. The materials consisted of (a) lesson plans to teach three behavior expectations, (b) a flow chart of intervention procedure with clear description of each step, (c) three behavior expectation posters, (d) a point chart for tracking team points, and (e) a table for reinforcer assessment (CW-FIT n.d.). The general education teacher implemented CW-FIT two times during math class one week before the experiment to get familiar with the CW-FIT procedures. After the first trial, the first author checked the intervention fidelity with the general education teacher and gave specific feedback accordingly (e.g. reminding the teacher to deliver positive feedback while giving team points). Notably, the first author found that students were confused about the definition of the three behavior expectations after being taught by the general education teacher. In response, the first author added a booster session to teach the three behavior expectations and introduce the game (i.e. CW-FIT Tier I) during math period before the experiment. Moreover, the first author conducted a class-wide reinforcer assessment and introduced the components of group contingency to the class, including setting up a goal at the beginning of the class, giving a point for each team following behavioral expectations, and rewarding a reinforcer to the teams meeting the goal. To clarify, the first author provided aforementioned reinforcers of group contingency in addition to the original reinforcement used by the general education teacher during intervention.

Self-management instruction: Tier II: The first author taught Ted with two typically developing students self-management skills in the resource room, which involved instructing them each to put a check mark on the self-management chart when showing one of the three behavior expectations. The instruction continued for nine sessions over three weeks (i.e. the last week of baseline and the first 2 weeks of B_1) until Ted could independently conduct self-management in the resource room, meaning Ted marked his behaviors with agreement that exceed 80% with the first author.

Measurement

Measurement and data analysis for dependent variables

The dependent variables were on-task and disruptive behaviors of Ted. We defined on-task behaviors as any appropriate behaviors related to class activities. This included active (e.g. raising his hand and answering questions) and passive (e.g. looking at the blackboard or materials for more than 5) student engagement behaviors. Disruptive behaviors were behaviors irrelevant to the class and interruptive; such as making noises, getting out of seat, playing or throwing objects, doodling, and talking to others without permission for more than 5 s.

Ten minutes after each class had begun, the first author employed a partial-interval recording procedure, documenting on-task and disruptive behaviors at any point during a 60-s interval for 20 min. The first author used a data sheet to record data. On-task and disruptive behaviors were recorded for each interval. If Ted presented any on-task or disruptive behaviors for more than 5 seconds within the interval, the first author would give a plus (+) on the data sheet whereas a minus (–) was for no occurrence of on-task or disruptive behaviors. Measures for dependent variables were calculated by dividing the total number of ''+'' by the total number of intervals, which was 20 in this study, and multiplying by 100.

The research team conducted visual inspection to evaluate the effects of CW-FIT on Ted's on-task and disruptive behaviors. The visual inspection provided the results of target behaviors within and across phases

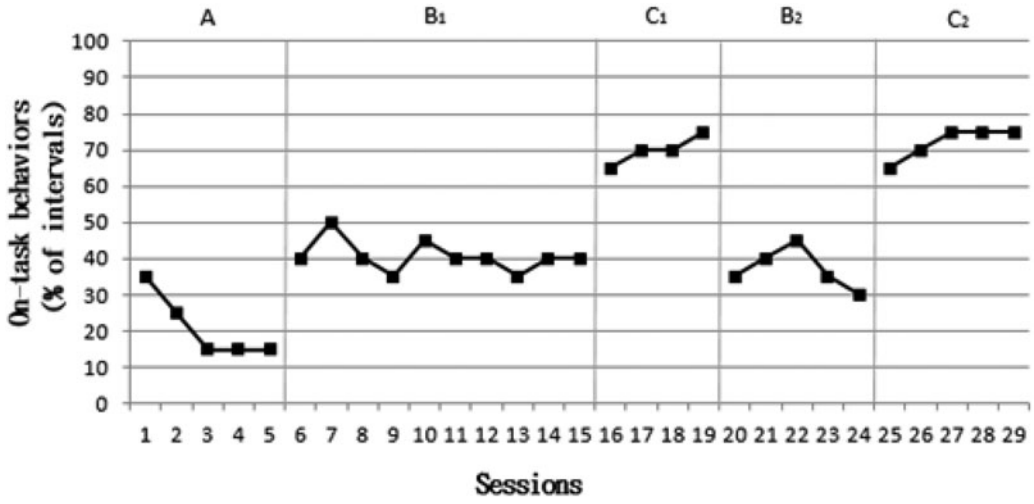


Figure 1 Percent of intervals of Ted's on-task behaviors

including (1) trend and trend stability, (2) level average, level difference, and level stability, and (3) percentage of overlapping data points across phases.

Inter-observer agreement

Before the experiment started, we examined the inter-observer agreement (IOA) between the first author and a second observer (i.e. a special education teacher at the same school). The first author provided training on the operational definitions for the target behaviors and the recording method, and two observers had in-depth discussion until their records reached an agreement above 80% for the IOA sessions. The IOA was calculated by dividing the total number of intervals with agreements by the total number of agreements and disagreements, and multiplying by 100 (Kennedy 2005). At the end of observer training, we found an average IOA of 91.5% with a range between 66 and 100% for on-task behaviors and an average IOA of 100% for disruptive behaviors.

During experiment, the research team examined IOA for 21% of the 29 data points. To ensure the stability of IOA throughout the experiment, the researchers checked the IOAs for two baseline sessions and four intervention sessions (including 2 sessions in B_1, 1 session in C_1, and 1 session in B_2) and all of the IOAs reached 100% for on-task and disruptive behaviors.

Procedural fidelity

We used two 12-item checklists of procedural fidelity to measure the degree to which the intervention was implemented as designed for Tier I and Tier II separately. The researchers marked ''yes'' if a procedure component was achieved and ''no'' in absence of the component. The procedural fidelity was calculated by dividing the total number of "yes" items by the total number of items on the checklists, and multiplying by 100. The procedural fidelity was measured on 24 sessions for Tier I intervention with an average score of 91.32% (range 83.33–100%) and nine sessions for Tier II intervention with an average score of 87.99% (range 75–91.67%). Both exceeded the predetermined score of 80%, indicating acceptable level of procedural fidelity (Kamps *et al.* 2015, Lower *et al.* 2016, Conklin *et al.* 2017).

Social validity

We used interviews and two questionnaires to assess social validity, one for the general education teacher and the other for the three peers on the same team with Ted. The questionnaire consisted of multiple-choice items and several open-ended questions about their perceptions on Ted's behavioral improvement and the procedure of CW-FIT. The multiple-choice items were on a 4-point Likert scale, with options ranging from "3 (great improvement)," "2 (apparent improvement)," "1 (slight improvement)," to "0 (no improvement)." The teacher questionnaire consisted of 10 items, including 5 items addressing the improvement of disruptive behaviors (e.g. playing objects, talking to others, getting out of seats during instruction, and etc.) and 5 about on-task behaviors (e.g. raising hand and answering questions, participating in team discussion, looking at the instructor, and etc.), with a possible score of 30. The student questionnaire consisted of 8 items of the improvement of Ted's disruptive behaviors with a total score of 24.

Results

On-task behaviors

The first research question of the present study was to investigate the effects of the CW-FIT on Ted's on-task behaviors. Figure 1 presented the results of on-task behaviors for Ted across five phases, including baseline (A), CW-FIT Tier I (B_1), Tier I + Tier II (C_1), Tier I (B_2), and Tier I + II (C_2).

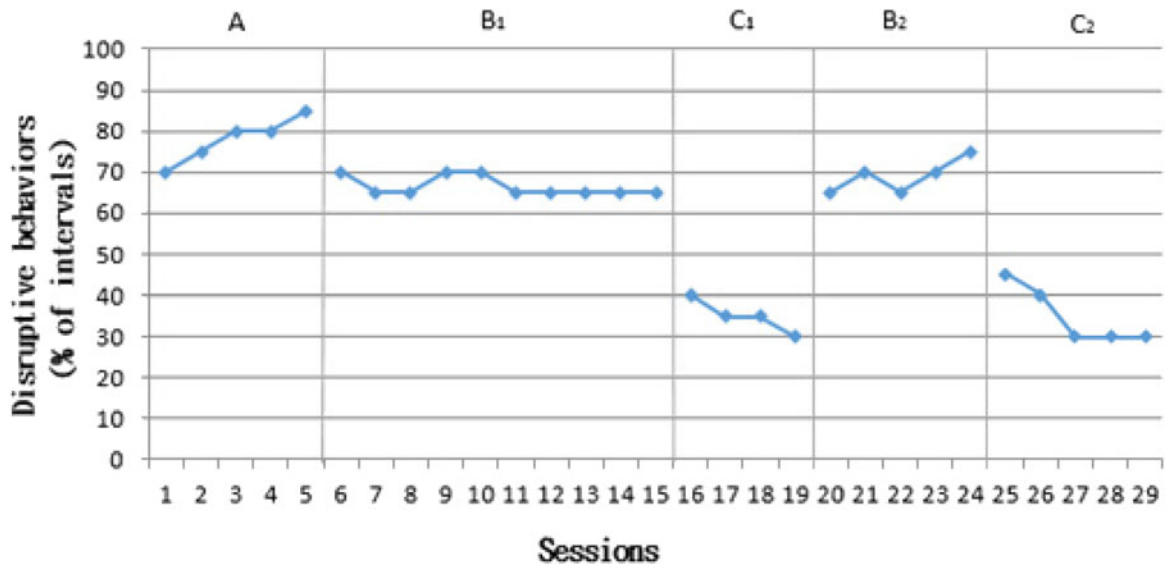


Figure 2 Percent of intervals of Ted's disruptive behaviors

During baseline (A), we recorded five data points. Ted's initial on-task behaviors were 35% of the 20 one-minute intervals and decreased to 15% with an average level of 21% (range 15–35%). After Ted showed stable on-task behaviors across three data points during baseline, we introduced Tier I (B_1) to the whole class and collected data on Ted's on-task behaviors. During B_1, we collected a total of 10 observational data points, and Ted's on-task behaviors increased to a mean of 40.5% of the intervals (range 35–50%). The change in level between the first data point of B_1 and the last data of the baseline was +25%, suggesting that Ted's on-task behaviors had an immediate increase from baseline (A) to intervention (B_1). The change in means across the two phases was from 21% to 40.5% with a 20% overlap in data points. The results showed that the Tier I was effective on improving the on-task behaviors but the overall level of improvement was insufficient [9, 10].

After the data showed a stable trend of on-task behaviors during B_1, we introduced the next phases, Tier I + Tier 2 (C_1) and recorded four observational data points during this phase. With the addition of self-management instruction, the results showed an improvement on Ted's on-task behaviors to a mean of 70% (range 65–75%). An immediate 25% increase of the level of Ted's on-task behaviors was observed from B_1 intervention to C_1. The change in means across the two phases was from 40.5 to 70% with 0% overlapping data points. Moreover, the results showed a change in trend from no clear trend to increase in Ted's on-task behaviors between the two phases (C_1/B_1). The results supported that the combination of Tier I and Tier II was more effective than Tier I only for increasing Ted's on-task behaviors.

Once the stability of data points was established, the condition was reversed to the Tier I (B_2) to validate the effects of Tier II. We collected five observational data points during B_2. Ted's on-task behaviors decreased to a mean of 37% (range 30–35%). We also found an immediate change in level of –40% when the condition was reversed from C_1 to B_2. The change in means across the two phases was from 70% to 37% with a 0% overlapping data points. Furthermore, a change in trend from increase to decrease in on-task behaviors between the two phases (B_2/C_1) was also observed. The results confirmed that the removal of Tier II instruction during this phase had an effect on the decrease of on-task behaviors.

An addition of self-management was re-introduced to Ted again (C_2) followed by the stable pattern was found in previous phase. During the final Tier I + Tier II phase, we collected five observational data points and found Ted's on-task behaviors increased to a mean of 72% (range 65–75%) After the teacher re-implemented self-management as a strategy, we observed an immediate 35% increase in the level of on-task behaviors from B_2 intervention to C_2. The change in means across the two phases was from 37% to 72% with 0% overlapping data points along with a change in trend from decrease to increase. The results confirmed that the combination of Tier I and Tier II was more effective than the Tier I only on the increase of the student with ASD in an inclusive classroom.

Disruptive behaviors

The second research question of this study was to investigate the effects of the CW-FIT on the operationally defined disruptive behaviors of tapping or chatting with peers, making noises, out-of-seat without permission, and playing with non-class relevant materials. Figure 2 summarized the experimental analyzes of five phases.

During baseline (A), the presence of Ted's disruptive behaviors ranged between 70 and 85% with a mean of

78% of the 20 one-minute intervals. Particularly, three out of the five baseline data points were above the mean average of 78% with a continuous increasing trend of disruptive behaviors. After the stability of the presentence of disruptive behaviors was established, the general education teacher implemented Tier I (B_1) to the whole class and the first author recorded Ted's behaviors to examine the effects. During B_1, the percentages of Ted's disruptive behaviors ranged between 65 and 70% with an overall average of 66.5%. Compared with the high rates in the baseline phase (70–85%), Ted's disruptive behaviors slightly decreased. As shown in Figure 2, an upward trend followed by a stable pattern could be visually inspected from baseline phase to B_1. The level difference between the first data point of B_1 and the last data of baseline was –15% with a 30% overlap in data points between the two phases. It showed that the whole class implementation of Tier I was slightly effective on improving Ted's disruptive behaviors. While Ted's disruptive behaviors began to decline, there remained a high rate of disruptive behaviors. Thus, the researchers introduced Tier I + Tier II (C_1) after observing a stable trend of disruptive behaviors.

In C_1, three trained students (including Ted) conducted self-management (Tier II) while the general education teacher implemented Tier I to the whole class. During C_1 the average percentage of Ted's disruptive behaviors was 35% of the 20 1-min intervals (ranging from 30 to 40% of intervals). A decline in the level of behaviors from B_1 to C_1 was observed, indicating a significant improvement in disruptive behaviors. More specifically, the level difference was –25% with 0% overlap in data and the average of occurrence of behaviors dropped from 66.5% (B_1) to 35% (C_1), and the disruptive behaviors have reduced substantially 43% from the baseline condition. The results altogether supported the combination of implementing both Tier I and Tier II CW-FIT instruction was more effective than Tier I only in decreasing Ted's disruptive behaviors. To validate the effects of Tier II on Ted's disruptive behaviors, we proceeded to a third intervention condition of reversal (B_2) once the stability of the data points in C_1 were established.

During B_2, the Tier II intervention, self-management, was withdrawn from the intervention condition and the general education teacher only implemented Tier I to the whole class. The average percentage of Ted's disruptive behaviors was 69% of all intervals (ranging from 65 to 75%). Among the five data points of this phase, three were above the average of 69%. Compared to the previous condition, the trend went from decline to flat, and the level of difference between the first data point of the B_2 and the last data of the C_1 was +35%, showing a negative change in data from C_1 to B_2. Moreover, the data also revealed that the average occurrence of Ted's disruptive behaviors increased from 30% (C_1) to 69% (B_2) with a 100% overlap of data points. The results suggest the removal of Tier II intervention during this phase had regressed the participant's classroom behaviors and decreased the effectiveness of the CW-FIT Tier intervention.

After the data showed stable pattern of disruptive behaviors in B_2, the teacher re-introduced self-management (C_2) to replicate the effects of Tier II intervention in addition to implementing Tier I (class-wide intervention). The average occurrences of Ted's disruptive behaviors was 35% of all intervals (ranging from 30 to 45%). The stability of the level and trend during C_2 suggested the effectiveness of the intervention on Ted's disruptive behaviors. Compared to B_2, the first data point of this phase and the last data point of B_2 showed a huge discrepancy (–30%) as well as the average occurrence of the disruptive behaviors of C_2 (35%) was lower than B_2 (69%) with 0% overlapping data point. In addition, the trend changed from flat to a decreasing when C_2 was implemented again. Overall, the data suggested that reintroducing the Tier I and II intervention showed great improvement of Ted's disruptive behaviors and supported the effectiveness of the CW-FIT intervention on the student with ASD in an inclusive education setting.

Social validity

The teacher's perception about CW-FIT: Of the 10-item questionnaire, the general education teacher rated 24 out of 30 points on the improvement of Ted's on-task and disruptive classroom behaviors. Based on the teacher's rating, Ted showed "great improvements (4 points)" on four out of five disruptive behaviors, including scrawling on textbooks, chatting with nearby classmates, getting out-of-seat without permission, and crying or shouting in class; "apparent improvement (3 points)" playing with materials. In the meantime, Ted also demonstrated "great improvement" on two out of five on-task behaviors, including responding to the teacher's commands immediately and following group activities. The teacher reported that Ted showed apparent improvement on participating group discussion and looking at teacher or blackboard; slight improvement on hand-raising. Specifically, the teacher stated that *"during group discussion, Ted was willing to sit with his teammates and paying attention to the group activities, though he did not initiate discussion*," and "*he used to not responding to the questions I asked but after the intervention he showed more willingness to respond to questions and instruction.*" In general, the teacher felt that CW-FIT showed effects on increasing Ted's on-task behaviors and decreasing his disruptive behaviors.

As to the implementation process, at first the general education teacher experienced some difficulties due to

unfamiliarity of the instructional steps of CW-FIT and sometimes missed the opportunity to deliver immediate feedback and reinforcement to the teams. After a few sessions, the teacher started to notice the behavioral change in all students. The general education teacher found that "*after implementing the CW-FIT intervention, I noticed that the students responded more actively and enjoyed class activities.*" During Tier I (class-wide intervention), the first author met with the general education teacher regularly to discuss the implementation procedures and collectively problem-solved challenges. The general education teacher also described that not only the student behaviors improved, but the class atmosphere become more positive as well.

Furthermore, the general education teacher also indicated willingness to continue implementing CW-FIT after completion of the experiment. The general education teacher stated that "*some at-risk students who were more easily to be distracted responded very well to the CW-FIT intervention, and showed progress in following directions and engaging in classroom activities*." Overall, the general education teacher observed the intervention's positive effects on Ted and the whole class, and provided the insight that at-risk students or students with social/emotional difficulties would benefit from the CW-FIT intervention.

Peers' perceptions about CW-FIT: Upon completion of the intervention, researchers also interviewed three peers on the same team with Ted and administered questionnaires regarding the effectiveness of the CW-FIT. With 24 points of possible scores, three peers rated 19, 20, and 21 respectively. All three teammates observed behavioral changes of Ted. Prior to the intervention, all three teammates indicated that Ted made noises, disrupted the class, left seat without permission, and was not engaging in team work all the time. Comments were as follow: "*he often took my stuff without asking my permission first,*" "*he would make disturbing noises during lecture,*" and "*he used to walk around the classroom in class and ignored the redirection to follow class rules.*" However, after the CW-FIT intervention, all three teammates reported Ted's improvement of behaviors and increased in participation. They found that "*he would sit quietly and made posters with other team members in the class,*" "*he made less noises during class and was paying more attention to the instruction*."

Moreover, the teammates and Ted were actively engaged in the whole-class instructional activities and expressed interests in the reinforcement and rewards upon the completion of team tasks. Another class-wide improvement was the strong sense of collectivism in which all members wanted to contribute and collaborate to earn rewards for their team. The three teammates all agreed that the level of engagement was increased and the class became more orderly after the intervention.

All in all, the general education teacher and the peers reported positive perceptions towards the CW-FIT intervention and recognized increased class participation on both individual and class-wide levels, which provides strong evidences for social validity of the intervention.

Discussion

The purpose of this study was to investigate the effect of CW-FIT Tier I (class-wide intervention) and Tier II (self-management) intervention on an elementary student with ASD in an inclusive classroom in Taiwan. This study expanded evidence for effectiveness of CW-FIT by providing data on a student with ASD in an Asian context. For the student with ASD, Tier I alone resulted in limited improvement on the student's on-task and disruptive behaviors which warranted implementation of the Tier II intervention. During Tier I + Tier II condition, the student with ASD suggested higher on-task behaviors and lower disruptive behaviors over baseline and when only Tier I is presented. These findings confirmed previous literature showing the effects of providing a targeted intervention for students with challenging behaviors in addition to primary level of class-wide intervention (Kamps *et al.* 2015, Wills *et al.* 2018).

Many variables contributed to the promising outcomes of this study. First, the first author (i.e. the resource room teacher) provided continuous support for general education teacher throughout the intervention. Specifically, the first author conducted a booster session on behavior expectations to the whole class and provided self-management training sessions that addressed learning needs of the student with ASD in resource room. The booster session and the pull-out training assured the target student understood the behavior expectations and could independently conduct self-management in inclusive classroom. In addition, the first author acted as an onsite coach for the general education teacher to provide feedback and emotional support for the general education teacher during intervention.

Second, during Tier I + Tier II conditions, the student with ASD and two other typical-developing peers conducted self-management while the general education teacher implemented class-wide CW-FIT intervention. From students' perspective, having typical-developing peers conducting self-management with Ted made Tier II components a privilege rather than a punishment and thus increase students' acceptance of the intervention (Kamps *et al.* 2015).

Third, feasibility of CW-FIT was particularly important for general education teachers given the burden keeping on tight teaching schedule for the examination-driven curriculum in Taiwan (Aldridge *et al.* 1999, Hsu and Chan 2008). State *et al.* (2017) also found that general education teachers favored with time-efficient and simple strategies. From the teacher's perspective, it was

feasible to add Tier II intervention to Tier I because all she had to do was giving those target students a specific feedback when rewarding team points, and it did not add much extra work for the teachers (Wills *et al.* 2010). Based on the fidelity and social validity data, the general education teacher could implement Tier I + Tier II intervention with ease after multiple practices showing that CW-FIT intervention is feasible in general education settings in Taiwan. The findings of this study confirmed that when teachers make efforts in providing students individualized and specific feedback within each 3- to 5-min interval during instruction, the students with ASD are more likely to remain on-task (Finn *et al.* 2015).

Limitations

While the results of this study showed promising effects of implementing CW-FIT Tier I and Tier II interventions in an inclusive classroom in Taiwan, several limitations should be considered. The limitations regarding research design are addressed as follow: First, we recruited students and teachers from only one inclusive classroom, which may limit generalizability of the conclusions. In particular, the general education teacher had a token economy system in place before the intervention, which may not be a standard practice among general education teachers in Taiwan. Second, the CW-FIT Tier I intervention was followed by Tier I + Tier II condition. Thus, it is unknown if self-management alone without class-wide intervention would have similar effects on the behaviors of the student with ASD. Third, the 2-week Tier I + Tier II phases were shorter than the 5-week Tier 1 phase. Finally, we did not collect maintenance data and thus the maintenance effects of CW-FIT require further investigation.

Other limitations regarding data collection were also notable. It was not clear to what extent the teacher employed praise and tokens before CW-FIT intervention without data collection on teacher behavior in baseline. Moreover, the researchers collected data solely on a student with ASD without systematically examining the behavior of the entire class, the teacher's behavior change, and the other two peers conducting self-management in this study. The intervals used to collect observational data (i.e. 60 s) were longer than what's been used in most intervention studies and thus might influence the results of observation. While the researchers documented social validity of CW-FIT from the teacher's and peers' perspectives, the target student, Ted, was not involved to examine his perception about class-wide and self-management intervention. Finally, while the researchers checked IOA for 21% of all data points, we did not examine data in the last phase due to time restraints. As the literature suggested checking IOA for 20% in all phases, this may hinder strength of the results (Kennedy 2005).

Conclusion

In conclusion, findings from this study support that the CW-FIT is a promising intervention in increasing on-task behaviors and decreasing disruptive behaviors of a student with ASD in an inclusive classroom in Taiwan. The effect of implementing multiple levels of CW-FIT was much more effective than implementing solely Tier I (class-wide intervention). The intervention was also well-received by the general education teacher and students.

The study provides promising evidence to prove that a general education teacher can work with resource room teachers to implement interventions to improve participation of students, including a student with ASD in an inclusive classroom. In particular, the general education teacher and resource room teachers can work together to establish concrete behavior expectations, provide behavioral instruction, teach self-management strategies systematically, and implement intervention on both individual and whole-class levels consistently. Future researchers should aim to expand settings/participants (e.g. teachers, typically developing peers, more students with ASD) and to extend the duration of the research.

References

Aldridge, J. M., Fraser, B. J. and Huang, T. C. I. 1999. Investigating classroom environments in Taiwan and Australia with multiple research methods. *The Journal of Educational Research*, 93(1), 48–62.

Caldarella, P., Williams, L., Hansen, B. D. and Wills, H. 2015. Managing student behavior with class-wide function-related intervention teams: an observational study in early elementary classrooms. *Early Childhood Education Journal*, 43(5), 357–365.

Chang, H. and Lin, H. 2011. 國小普通班自閉症學生學校適應與學校支持之研究-以中部地區爲例 [The study of the school adjustment and the school support system for the autistic elementary school students of regular classes in central Taiwan area]. *Bulletin of Special Education and Rehabilitation*, 25, 25–46.

Chen, P. and Chiu, C. 2015. Inclusive education for students with low-incidence disabilities in Taiwan – where we are, what we have learned. In E. West, ed. *International perspectives on inclusive education.* Bradford, UK: Emerald, pp. 203–219.

Chiang, Y. 2011. 「不想成爲班上的小白兔！」：以符號詮釋自閉症學童在融合教育中友誼建立之挑戰 ["Don't want to be a pet rabbit in class!": using symbolic interpretation to explore the challenges of friendship building on students with autism in inclusive education]. *Bulletin of Special Education*, 36, 87–114.

Chiu, S. 2001. 普通班教師對特殊需求學生之因應措施、所面對之困境以及所需之支持系統 [*The coping strategies, barriers, and the supporting needs for the regular classroom teachers with teaching special students*]. *Bulletin of Special Education,* 21, 1–26. Retrieved from: http://bse.spe.ntnu.edu.tw/upload/journal/prog/K1N_1841_320M_247Y1023.pdf

Conklin, C. G., Kamps, D. and Wills, H. 2017. The effects of class-wide function-related intervention teams (CW-FIT) on students' prosocial classroom behaviors. *Journal of Behavioral Education*, 26(1), 75–100.

CW-FIT. n.d. CW-FIT Archived [Webpage]. Retrieved from: https://cwfit.ku.edu/cw-fit-%E5%9C%8B%E8%AA%9E%E7%BF%BB%E8%AD%AF

De Boer, A., Pijl, S. J. and Minnaert, A. 2011. Regular primary school teachers' attitudes towards inclusive education: a review of the literature. *International Journal of Inclusive Education*, 15(3), 331–353.

Educational Fundamental Act. 2013. Retrieved from: https://law.moj.gov.tw/Eng/LawClass/LawContent.aspx?PCODE=H0020045

Enforcement Rules of the Special Education Act. 2013. Retrieved from: https://law.moj.gov.tw/Eng/LawClass/LawContent.aspx?PCODE=H0080032

Ferguson, D. L. 2008. International trends in inclusive education: the continuing challenge to teach each one and everyone. *European Journal of Special Needs Education*, 23(2), 109–120.

Forlin, C. and Chambers, D. 2011. Teacher preparation for inclusive education: increasing knowledge but raising concerns. *Asia-Pacific Journal of Teacher Education*, 39(1), 17–32.

Finn, L., Ramasamy, R., Dukes, C. and Scott, J. 2015. Using watch minder to increase the on-task behavior of students with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 45(5), 1408–1418.

Hsu, C. and Chan, S. 2008. 國小教師對不同類別身心障礙學生就讀普通班意見之調查研究 [A survey study on aspects of elementary school teachers toward students with different disabilities in regular classes]. *Bulletin of Special Education Research*, 19, 25–49.

Huang, H. and Lan, W. 2012. 普通班融合教育情境中對自閉症學生之教學策略 [Teaching strategies for students with autism in general education class]. *The Elementary Education Journal*, 59(4), 15–25.

Jolstead, K. A., Caldarella, P., Hansen, B., Korth, B. B., Williams, L. and Kamps, D. 2017. Implementing positive behavior support in preschools: an exploratory study. *Journal of Positive Behavior Interventions*, 19(1), 48–60.

Kamps, D., Conklin, C. and Wills, H. 2015. Use of self-management with the CW-FIT group contingency program. *Education and Treatment of Children*, 38(1), 1–32.

Kamps, D., Wills, H. P., Heitzman-Powell, L., Laylin, J., Szoke, C., Petrillo,T., and Culey, A. 2011. Class-wide function-related intervention teams: effects of group contingency programs in urban classrooms. *Journal of Positive Behavior Interventions*, 13(3), 154–167.

Kennedy, C. H. 2005. *Single-case designs for educational research*. New York: Pearson.

Lai, H. R., Chou, W.-L., Miao, N.-F., Wu, Y.-P., Lee, P.-H., and Jwo, J.-C. 2015. A comparison of actual and preferred classroom environments as perceived by middle school students. *Journal of School Health*, 85(6), 388–397.

Lin, P. 2011. 特殊教育並不特殊：從特殊教育的特殊性談融合教育之現況與未來 [Special education isn't special: the current and future special education]. *Special Education Quarterly, 120*, 11–18. Retrieved from: http://www.airitilibrary.com/Publication/alDetailedMesh?docid=P20110718004-201109-201206200024-201206200024-11-18%2B52

Lower, A., Young, K. R., Christensen, L., Caldarella, P., Williams, L. and Wills, H. 2016. Effects of a Tier 3 self-management intervention implemented with and without treatment integrity. *Education and Treatment of Children*, 39(4), 493–520.

Marzano, R. 2013. *Classroom management that works: research-based strategies for every teacher*. Alexandria, VA: ASCD.

Ministry of Education. 2017. 一〇六年度特殊教育統計年報. [*Annual special education statistical report for 106 school year*]. Taipei, Taiwan: Ministry of Education.

Naylor, A. S., Kamps, D. and Wills, H. 2018. The effects of the CW-FIT group contingency on class-wide and individual behavior in an urban first grade classroom. *Education and Treatment of Children*, 41(1), 1–30.

Pivik, J., McComas, J. and Laflamme, M. 2002. Barriers and facilitators to inclusive education. *Exceptional Children*, 69(1), 97–107.

Special Education Act. 2014. Retrieved from: https://law.moj.gov.tw/Eng/LawClass/LawContent.aspx?PCODE=H0080027

State, T. M., Harrison, J. R., Kern, L. and Lewis, T. J. 2017. Feasibility and acceptability of classroom-based interventions for students with emotional/behavioral challenges at the high school level. *Journal of Positive Behavior Interventions*, 19(1), 26–36.

Sun, S. and Huang, T. 2018. 普通班身心障礙學生學校支持服務模式之建構 [Establishing a model of school support services for students with disabilities in general education classroom]. *Contemporary Educational Research Quarterly*, 23(1), 77–113.

UN Committee on the Rights of Persons with Disabilities (CRPD), General Comment. 2016. No. 4 Article 24: right to inclusive education. Retrieved from: https://www.ohchr.org/Documents/…/CRPD/GC/RighttoEducation/CRPD-C-GC-4.doc

Wills, H., Iwaszuk, W. M., Kamps, D., and Shumate, E. 2014. CW-FIT: group contingency effects across the day. *Education and Treatment of Children*, 37(2), 191–210.

Wills, H., Kamps, D., Fleming, K. and Hansen, B. 2016. Student and teacher outcomes of the Class-Wide Function-Related Intervention Team efficacy trial. *Exceptional Children*, 83(1), 58–76.

Wills, H., Wehby, J., Caldarella, P., Kamps, D. and Romine, R. S. 2018. Classroom management that works: a replication trial of the CW-FIT. *Exceptional Children*, 84(4), 437–456.